POKHARA UNIVERSITY

Level: Bachelor | Semester: Spring | Year : 2023
Programme: BE | Full Marks: 100
Course: Digital Signal Analysis and Processing | Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

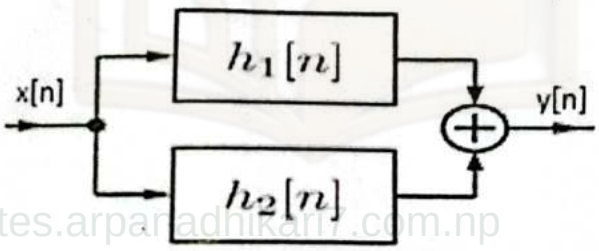
The figures in the margin indicate full marks.

Attempt all the questions.

1. a) A digital communication link carries binary coded words representing samples of an input signal $x_a(t) = 5\cos 600\pi t + 7\cos 800\pi t$. The link is operated at 1000 bits/sec and each input sample is quantized into 1024 different voltage levels. 7

   i. What is the sampling frequency and folding frequency?
   ii. What is the Nyquist rate for the signal $x_a(t)$?
   iii. What are the frequencies in the resulting discrete time signal x[n]?
   iv. What is aliasing effect?

   b) Two subsystems $h_1[n]$ and $h_2[n]$ are interconnected as shown in the block diagram. Determine the response of the system if; $h_1[n] = \{1,4,2\}$ and $h_2[n] = \{2, 3\ 1\}$, when excited by input $x[n] = \{2, 4\}$. 8



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2. a) Determine which signals are periodic and compute fundamental period. 8

   i. $\cos\left(\frac{\pi n^2}{8}\right)$ ii. $\sin\left(\frac{3\pi}{2}\right)\cos\left(\frac{5\pi n}{4}\right)$

   b) Define z-transform and Region of Convergence. Find z-transform of following signal $x[n] = a^n u[n]$. 7

3. a) Why do we need DFT when we have DTFT? Determine the circular convolution of the sequence: $x_1(n) = \{4,3,2,1\}$ and $x_2(n) = \{8,7,6,5\}$ 7

   b) Determine the direct form realization of the following difference equation. 8

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$2y(n) + y(n-1) - 4y(n-3) = x(n) + 3x(n-1)$

Compare FIR and IIR

4. a) Determine the cascade and parallel realization of the discrete time systemdescribed by differential equation. 8

$$y[n] = -\frac{3}{4}\, y[n-1] + \tfrac{1}{4}\, y[n-2] + x[n] + \frac{1}{2}\, x[n-1]$$

b) Obtain the lattice ladder structure of the discrete time system described bythe differential equation 7

$$y[n] = -\frac{3}{4}\, y[n-1] + \tfrac{1}{4}\, y[n-2] + x[n] + \frac{1}{2}\, x[n-1]$$

Also check the stability of the filter

5. a) Design a digital low pass Butterworth filter by applying bilinear transformation technique for the given specifications. 7

Pass band edge = 120Hz

Pass band attenuation = 1dB

Stop band edge = 170Hz

Stop band attenuation = 16 dB Assume sampling frequency of 512 Hz

b) Obtain $H(z)$ using the impulse invariant techniques for an analog system function which is given by: 8

$$H_a(S) = \frac{1}{(S+0.5)(S^2+0.5S+2)}$$

6. a) Design a linear FIR filter using Kaiser window to meet the following specifications: 8

$0.99 \le |H(e^{\wedge}jw| \le 1.01$; for $0 \le |w| \le 0.19\pi$

$|H(e^{\wedge}jw| \le 0.01$; for $0.21\pi \le |w| \le \pi$

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b) By using Hanning window, design a low pass filter to approximate the ideal response given by: 7

$$H(e^{jw}) = \begin{cases} 1 & for \quad -\frac{\pi}{6} \le w \le \frac{\pi}{6} \\ 0 & otherwise \end{cases}$$

Use the filter length of M=9 for your design

7. Write short notes on: **(Any two)** 2×5

a) Sampling and quantization of analog signal,

b) BIBO Stable

c) Frequency Sampling

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